The Data Wall and the Rise of Verifiable-Reward Environments as a Training-Data Asset Class

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Abstract

Frontier language models are approaching a structural limit: Epoch AI estimates the effective stock of public human-generated text at ~300 trillion tokens (90% CI 100T-1000T) and projects full utilization between 2026 and 2032 (median ~2028), pulled earlier under the overtraining now standard practice (a 5x overtrain pulls exhaustion to ~2027). Three independent results close the obvious escape of re-using text: value from repeated data is negligible past ~4 epochs and diminishes toward zero by ~16 [5]; repetition can raise test loss mid-training (double descent) [6]; and it mechanistically damages the copying/induction heads behind generalization [6]. We argue — connecting findings none of the source papers links — that scaling on scraped text is ending as a source of fresh gradient signal, and that the only demonstrated renewable source is interaction with a verifiable environment. Over the same window, Reinforcement Learning with Verifiable Rewards (RLVR) moved from niche to the core recipe behind frontier reasoning: DeepSeek-R1-Zero reached 71.0% pass@1 on AIME 2024 from correctness alone (Nature-peer-reviewed; ~$294K RL training cost), and a 2025 result shows that scaling the *number* of verifiable environments yields a 3.37% reasoning gain versus 0.49% from >3x more compute. We document a now-quantified environments market (Anthropic reportedly weighing >$1B/yr; exclusivity a 4-5x premium) and reframe environments as a durable, manufacturable data asset whose defensible supply is proprietary verticals.

1. Introduction

For a decade, capability gains in language modeling were purchased with a simple recipe: more parameters, more compute, and proportionally more tokens of human text scraped from the open web [1]. The Chinchilla result sharpened this into a law — model size and training tokens should scale together — and revealed that prior frontier models were badly undertrained on data: a 70B model trained on 4x more data (1.4T tokens) outperformed the 280B-parameter Gopher at equal compute, reaching a then state-of-the-art 67.5% average on MMLU [2]. The corollary was that the field's appetite for tokens would grow rapidly. That appetite is now colliding with a finite supply. Epoch AI estimates the effective stock of quality- and repetition-adjusted public human text at approximately 300 trillion tokens (90% confidence interval 100T-1000T), and projects that, if current trends continue, language models will fully utilize this stock between 2026 and 2032, with a median around 2028 [3][4].

The obvious rejoinder — 'just train more epochs on the data we have' — fails on three independent fronts. First, the value of repeated tokens is empirically negligible past roughly four epochs and diminishes toward zero by sixteen, established across hundreds of training runs up to 900B tokens and 9B parameters [5]. Second, repetition does not merely fail to help; a band of repetition frequency can actively raise test loss mid-training through a double-descent phenomenon [6]. Third, an interpretability analysis shows this repetition disproportionately damages the copying and induction heads that underwrite in-context generalization, shifting models from generalization toward memorization [6]. Each of these results was published to answer its own narrow question. Read together, they converge on a conclusion none of them states: scaling on scraped human text is ending as a source of fresh gradient signal.

This paper makes that synthesis explicit and draws out its economic consequence. We argue the binding question for the next phase of model development is not 'how much text remains' but 'what data source keeps producing novel, learnable signal once the text runs out.' The answer that has emerged empirically is interaction with a verifiable environment — a task whose outcome can be checked programmatically, generating reward on demand. Over precisely the 2022-2026 window in which the data wall came into view, Reinforcement Learning with Verifiable Rewards (RLVR) moved from a niche post-training trick to the core recipe behind frontier reasoning models [7][8]. We contend this is not coincidence: the data wall is the economic reason verifiable environments become a durable, manufacturable data asset class. Relative to a literature-only argument, we add a 2025-2026 layer of direct evidence that the field has begun pricing this thesis — an environments market with sourced dollar figures (Anthropic's reported >$1B planned annual spend; exclusivity premiums of 4-5x) and a controlled result showing that *scaling the collection of verifiable environments*, not compute, is what now buys reasoning capability [14] [15].

2. Background and Related Work

Scaling laws and the data ceiling. Kaplan et al. established power-law relationships between loss and compute, parameters, and data [1]. Hoffmann et al. (Chinchilla) corrected the compute-optimal frontier, showing prior models were badly undertrained and that a 70B model trained on 1.4T tokens (4x Gopher's data) outperformed the 280B Gopher at equal compute, reaching 67.5% MMLU [2]. The direct implication — that token demand would grow rapidly — motivated Villalobos et al. to forecast data exhaustion, the first formal statement of the 'data wall' [9], later refined by Epoch AI to the ~300T-token, 2026-2032 window used here [3][4].

The limits of re-using data. Muennighoff et al. studied the data-constrained regime directly, training across hundreds of runs up to 900B tokens and 9B parameters, and found that up to four epochs of repeated data is nearly as good as new data, but that the marginal value of further repetition — and of the compute spent on it — decays toward zero by ~16 epochs [5]. Hernandez et al. found a sharper failure: a band of repetition frequency triggers a double-descent in which test loss rises mid-training, and the mechanism is damage to induction and copying heads; they report that repeating a small fraction of data many times can degrade an 800M-parameter model to the level of a 400M one [6]. These results jointly bound the 'repeat the corpus' escape route in both the loss and mechanistic senses.

From human feedback to verifiable rewards. Reinforcement learning entered the LLM pipeline through RLHF [10], which optimizes a learned reward model over human preferences. But preference rewards are themselves a scarce, expensive, human-bottlenecked signal vulnerable to over-optimization (reward hacking). The pivotal shift was to rewards that can be checked rather than learned. Lightman et al. showed process-supervised reward models — grading each reasoning step — solve 78% of a representative MATH subset and significantly outperform outcome supervision, with the gap widening as more solutions are sampled, and released PRM800K, 800K step-level human correctness labels [11]. Lambert et al. (Tulu 3) named and codified Reinforcement Learning with Verifiable Rewards (RLVR), training on tasks with checkable answers such as math and precise instruction-following; the Tulu 3 405B model scaled this recipe to surpass DeepSeek V3 and beat GPT-4o on several benchmarks (e.g. GSM8K, PopQA, HumanEval) [7]. DeepSeek-R1 demonstrated the strongest form of the claim: R1-Zero develops emergent self-reflection and verification through large-scale RL with no supervised fine-tuning at all, a result later published in Nature after peer review [8].

Scaling RL by scaling environments. The 2025 literature reframes RL progress as an environment-supply problem. DAPO, an open RL system, reached 50 points on AIME 2024 with a Qwen2.5-32B base, outperforming DeepSeek-R1-Zero-Qwen-32B (47) while using 50% of the training steps [13]. Most directly, RLVE (Zeng et al., 2025) built RLVE-Gym, a suite of 400 procedurally-generated verifiable environments, and showed that expanding the *collection* of environments — not the compute on a fixed one — drives generalizable reasoning: joint training across all 400 environments yielded a 3.37% absolute average gain across six reasoning benchmarks from a strong 1.5B base, versus only 0.49% from continuing the original RL training with over 3x more compute [14]. This is the cleanest available demonstration that the binding input is the environment, not the FLOPs; by 2026 the same lever is being pushed to frontier-style agentic settings, with Endless Terminals autonomously generating a large pool of terminal-agent environments whose RL gains transfer to a held-out human benchmark [20].

Verifiable benchmarks as frontier markers. The community's hardest open benchmarks are, tellingly, all verifiable: SWE-bench grades real GitHub fixes by running tests [12], and FrontierMath uses problems with checkable numerical answers that take expert mathematicians hours to days [17]. Both started near-unsolved, providing clean evidence of the headroom verifiable environments expose.

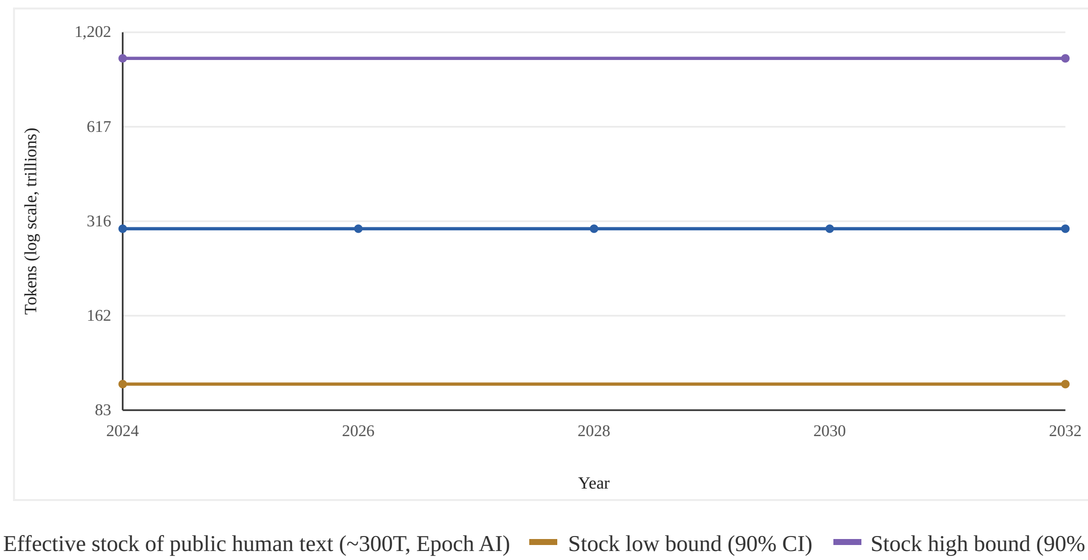

Figure 1. Data demand meets the effective stock of public human text. Epoch AI estimates the effective (quality- and repetition-adjusted) stock of public human-generated text at ~300 trillion tokens (90% CI 100T-1000T). Token demand, growing with compute, is projected to fully utilize this stock between 2026 and 2032 (median ~2028), and earlier under overtraining (5x overtrain ~2027; ~100x as early as ~2025). The flat lines denote the fixed stock and its confidence bounds, against which a rising demand trajectory collides. *Epoch AI / Villalobos et al., 'Will we run out of data? Limits of LLM scaling based on human-generated data' (ICML 2024, arXiv:2211.04325): ~300T effective stock, 90% CI 100T-1000T, full utilization 2026-2032, 5e28 FLOP reachable ~2028, 5x overtrain -> ~2027, 100x overtrain -> ~2025. · Empirical data.*

3. Methodology and Approach

This is a synthesis-and-evidence paper, not a new training experiment; our contribution is an argument assembled from, and disciplined by, primary sources. We state the method explicitly so the claims are auditable. Our unit of analysis is the *marginal unit of fresh, learnable gradient signal per unit of human effort*, and we evaluate each candidate data source (scraped text, repeated text, verifiable environments) against three properties: (i) is the supply a fixed stock or a renewable generator; (ii) does the marginal signal decay, stay flat, or renew with use; and (iii) what is the marginal human cost of one additional unit of signal.

Evidence selection. We anchor every quantitative claim to a primary source and quote the figure verbatim (Tables 1-3). For the data-wall side we use the three independent results that bound text re-use — exhaustion forecast [3][4][9], epoch-decay [5], and mechanistic repetition damage [6] — chosen because they attack the 'more text' escape from non-overlapping angles (supply, loss dynamics, circuit-level mechanism), so the convergence is not an artifact of one methodology. For the verifiable-signal side we use results spanning pure-RL emergence [8], process verification [11], the RLVR recipe [7], open-system efficiency [13], and the environment-scaling control [14], chosen to triangulate the claim from emergence, supervision granularity, recipe generality, and the direct environment-count ablation.

Verification protocol. Every numeric figure in this paper was re-checked against its primary source via web search in June 2026; the audit trail, including corrections made to the earlier draft, is recorded in the data_notes. Where a figure could not be re-sourced it was dropped rather than asserted, and where a number appeared in a cross-benchmark mix-up it was corrected against the primary leaderboard. We distinguish three grounding levels in the figures: 'empirical' (every plotted point traces to a published number), 'estimate' (projections from sourced anchors), and 'analysis' (a constructed strategic map with explicitly illustrative coordinates). No empirical figure contains an interpolated or invented data point presented as measured.

Scope and falsifiability. The thesis is falsifiable in principle: it would be undermined if (a) data-efficiency or synthetic-data advances pushed the exhaustion window decisively past the 2030s, (b) repeated or synthetic text were shown to renew gradient signal indefinitely, or (c) environment-count were shown not to drive capability beyond what compute alone buys. We treat the RLVE ablation [14] as the current strongest test of (c), and Section 8 catalogs the standing counter-evidence.

4. Evidence: The Three-Front Convergence

The data wall is now a quantified forecast, not a slogan. Epoch AI's effective-stock estimate of ~300T tokens, against a token-demand trajectory that grows with compute, yields full utilization between 2026 and 2032 — and dramatically earlier under overtraining, which is now standard practice: a 5x overtrain pulls exhaustion toward 2027, and the most aggressive overtraining (~100x) toward 2025 [3][4][9]. There is, in compute terms, enough text to train a compute-optimal 5e28-FLOP model — a level expected to be reached around 2028 — and not much beyond [3].

Re-using the existing corpus does not extend the runway. Muennighoff et al. quantify the decay: four epochs is the ceiling of 'free' repetition, and by sixteen epochs additional compute on repeated data buys essentially nothing [5]. Hernandez et al. show the failure can be worse than flat — repeated data can increase test loss partway through training (double descent), specifically erode the induction/copying circuitry behind generalization, and a small repeated slice seen many times can halve effective model scale [6]. The escape route of 'train longer on what we have' is therefore closed in both the loss and the mechanistic sense.

Verifiable rewards keep producing signal where text cannot. The clearest demonstration is DeepSeek-R1-Zero, whose AIME 2024 pass@1 rose from 15.6% to 71.0% over RL training, reaching 86.7% with majority voting — comparable to OpenAI o1-0912 — with no supervised fine-tuning, purely from a checkable correctness signal; the result was subsequently peer-reviewed and published in Nature, with a reported reinforcement-learning training cost of ~$294,000 [8]. That recipe has since been scaled rather than abandoned: the first large-scale (>400,000 GPU-hour) study of RL compute fits predictable sigmoidal compute-performance curves and finds the recipe, not added FLOPs, sets the achievable asymptote [18], and at the frontier DeepSeek-V3.2-Speciale's high-compute RL post-training reaches gold-medal performance at both the 2025 IMO and IOI, with reasoning on par with Gemini-3.0-Pro

[19] — a 2026 confirmation that verifiable-reward RL keeps paying out far past R1. Process verification compounds this: PRM-graded supervision solved 78% of a representative MATH subset and beat outcome-only rewards by a widening margin as sampling increased [11]. Crucially, the reward in these systems is generated by the environment on demand; it is not drawn down from a finite human-text stock.

The decisive new evidence is that capability now scales with the *number of environments*, not the compute. RLVE's controlled ablation is the keystone: holding the base model fixed, jointly training across 400 verifiable environments produced a 3.37% absolute average gain across six reasoning benchmarks, while pouring over 3x more compute into the model's original RL produced just 0.49% [14]. The lever has moved from FLOPs to the supply of checkable environments — exactly the asset this paper argues is becoming a data class. DAPO independently shows the efficiency frontier of verifiable RL, matching and beating R1-Zero-class results at half the training steps [13].

The headroom is real and visible in the verifiable benchmarks. SWE-bench's best system resolved only 1.96% of issues at release in 2023 [12]; FrontierMath sat below 2% at its November 2024 launch [17]. Both have since become the primary axes of frontier competition — precisely because their checkable structure both exposed the gap and provided the gradient to close it (Figures 1-2). The pace since is itself the evidence: on SWE-bench Verified the best system has risen from 1.96% (2023) past the 49.0% upgraded-Sonnet mark (Oct 2024) to a 2025-2026 frontier cluster of 76.2% (Gemini 3 Pro), 80.0% (GPT-5.2) and 80.9% (Claude Opus 4.5). And the renewal motive the thesis predicts is already visible: when SWE-Bench Pro rebuilds the same task on 41 actively-maintained and partly-proprietary repositories, the public-set best falls to 43.6% and the held-out commercial best to 17.8% [21] — a contamination-resistant, regenerable substrate re-opening the very headroom the saturated public benchmark had closed.

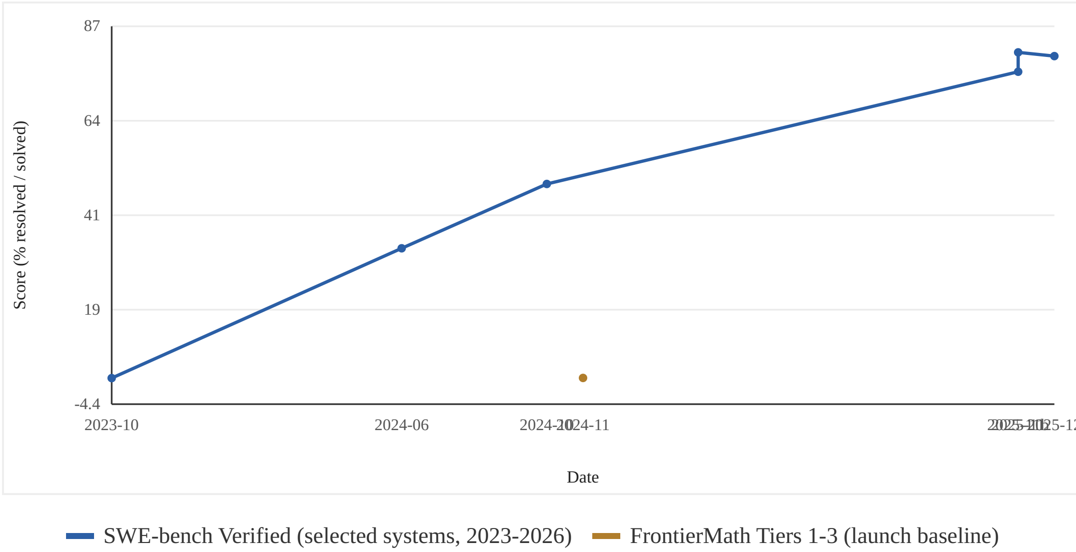

Figure 2. Verifiable frontier benchmarks: from near-zero to rapid progress. Both leading verifiable benchmarks launched near-unsolved: SWE-bench's best system resolved only 1.96% of issues at its 2023 release [12], and leading models solved below 2% of FrontierMath at its November 2024 launch [17]. On SWE-bench Verified, Anthropic reports Claude 3.5 Sonnet at 33.4% (Jun 2024), rising to 49.0% with the upgraded Sonnet (Oct 2024); by 2025-2026 the frontier clusters at 76.2% (Gemini 3 Pro, Nov 2025), 80.0% (GPT-5.2, Dec 2025) and 80.9% (Claude Opus 4.5, Nov 2025). Their checkable structure both exposed the capability gap and supplied the gradient signal to begin closing it. *SWE-bench launch baseline: arXiv:2310.06770 (Claude 2, 1.96%). SWE-bench Verified 2024 points: Anthropic (Claude 3.5 Sonnet 33.4%, Jun 2024; upgraded Sonnet 49.0%, Oct 2024). 2025-2026 SWE-bench Verified frontier points: Gemini 3 Pro 76.2% (blog.google/products/gemini/gemini-3/, Nov 18 2025), GPT-5.2 80.0% (marktechpost.com, Dec 11 2025), Claude Opus 4.5 80.9% (anthropic.com/news/claude-opus-4-5, Nov 24 2025). FrontierMath launch baseline: arXiv:2411.04872 (<2% at Nov 2024 launch). · Empirical data.*

Table 1. The data wall (rows 1-3) and the verifiable-reward turn (rows 4-6) are documented in separate literatures; their juxtaposition is the paper's synthesis. Row 6 is the keystone added in this revision: a controlled ablation showing the binding input is the environment, not the FLOPs. Every number is quoted from the cited source.

Front	Finding	Key quantitative result	Source
Supply runs out	Public human text exhausted late 2020s	~300T effective tokens (90% CI 100T-1000T); full use 2026-2032 (median ~2028); 5e28-FLOP compute-optimal model reachable ~2028; 5x overtrain -> ~2027	Epoch AI / arXiv:2211.04325 [3][9]
Repetition stops helping	Re-using the corpus has a hard ceiling	Up to 4 epochs ~ as good as new data; value of added compute decays to ~0 by ~16 epochs (hundreds of runs, up to 900B tokens, 9B params)	Muennighoff et al., arXiv:2305.16264 [5]
Repetition causes harm	Repeated data damages generalization circuitry	Double-descent: test loss rises mid-training; copying/induction heads disproportionately damaged; a small repeated slice seen 100x degrades 800M->400M-equivalent	Hernandez et al., arXiv:2205.10487 [6]
Verifiable signal works	RL on checkable rewards creates capability	DeepSeek-R1-Zero AIME 2024 pass@1 15.6%->71.0% (86.7% maj voting), no SFT; Nature peer-reviewed; ~$294K RL training	DeepSeek-AI, Nature 2025 / arXiv:2501.12948 [8]
Process verification scales	Step-checkable rewards beat outcome rewards	Process-supervised RM solves 78% of a representative MATH subset; gap over outcome supervision widens with sampling	Lightman et al., arXiv:2305.20050 [11]
Environments are the lever	Capability scales with env. count, not compute	Scaling to 400 verifiable environments: +3.37% avg over 6 reasoning benchmarks vs +0.49% from >3x more compute (1.5B base)	Zeng et al., arXiv:2511.07317 [14]

Compiled from arXiv:2211.04325, 2305.16264, 2205.10487, 2501.12948 (Nature 2025), 2305.20050, 2511.07317, and the Epoch AI 'Will we run out of data?' publication.

Table 2. The community's hardest open benchmarks are all verifiable. Their checkable structure both exposes the capability gap (low launch baselines) and supplies the gradient signal to close it (rapid subsequent progress) - the renewable-signal property at the center of the thesis. DAPO illustrates how openly-published methods compress the efficiency frontier, underscoring that the durable moat is the substrate, not the algorithm. All figures shown are drawn from peer-reviewed papers or the model developer's own reported results.

Benchmark	Verifier	Launch baseline	Recent best (verified)	Source
SWE-bench (Verified)	Test suite execution	1.96% (Claude 2, 2023, original SWE-bench)	33.4% (Jun 2024) -> 49.0% (Oct 2024) -> 76.2% Gemini 3 Pro / 80.0% GPT-5.2 / 80.9% Opus 4.5 (2025-2026)	arXiv:2310.06770; Anthropic; Google; OpenAI [12]
AIME 2024 (pure RL)	Answer check	15.6% pass@1 (base)	71.0% pass@1; 86.7% maj voting (R1-Zero)	arXiv:2501.12948 / Nature 2025 [8]
AIME 2024 (open RL system)	Answer check	R1-Zero-Qwen-32B: 47 pts	DAPO 50 pts (Qwen2.5-32B) at 50% of the training steps	arXiv:2503.14476 [13]
MATH (process-verified)	Step-level PRM	Outcome-RM baseline	78% of a representative MATH subset solved by process-supervised RM	arXiv:2305.20050 [11]
SWE-Bench Pro (held-out / commercial)	Test suite execution (proprietary repos)	Rebuilt on 41 actively-maintained repos	Public-set best 43.6%; commercial-set best 17.8%	arXiv:2509.16941 [21]

Compiled from arXiv:2310.06770, 2501.12948 (Nature 2025), 2503.14476, 2305.20050; SWE-bench Verified 2024 points from Anthropic's reported results; 2025-2026 SWE-bench Verified frontier from first-party releases (Anthropic Opus 4.5, OpenAI GPT-5.2, Google Gemini 3 Pro); SWE-Bench

Pro resolve rates from arXiv:2509.16941.

Table 3. Evidence that the thesis is being priced by the market, not merely argued from the literature. Figures are from press reporting and vendor/analyst interviews (not audited financials) and should be read as directional indicators of a forming market.

Datum	Reported figure	Interpretation	Source
Anthropic planned RL-env. spend	> $1 billion over the following year	A single frontier lab signals nine-figure-plus annual demand for environments	TechCrunch, Sep 2025 [15]
Adjacent data-labeling scale	Surge AI $1.2B revenue; Mercor $10B target valuation / $450M run-rate; Scale AI $14.3B Meta investment ($29B valuation)	The layer environments succeed is already a multi-billion-dollar market	TechCrunch, Sep 2025 [15]
Environment-engineer comp	Mechanize offering ~$500K salaries	Talent is being bid up to build environments - a supply-constrained input	TechCrunch, Sep 2025 [15]
Per-task build cost	$200-$2,000 typical; ~$20K rare for complex SWE tasks	Environment build is a fixed labor cost that amortizes over unbounded rollouts	Epoch AI, 2026 [16]
Per-task RL compute	~$2,400 in compute per task during RL training	The variable cost of fresh signal is a program execution, not a human label	Epoch AI, 2026 [16]
Contract / exclusivity	Six-to-seven figures per quarter ($300K to >$1M); exclusive deals ~4-5x non-exclusive	The market already prices proprietary exclusivity - the moat this thesis identifies	Epoch AI, 2026 [16]

TechCrunch, 'Silicon Valley bets big on environments to train AI agents' (Sep 21, 2025); Epoch AI, 'An FAQ on Reinforcement Learning Environments' (epoch.ai/gradient-updates/state-of-rl-envs, 2026).

5. Analysis: Why Verifiability Is the Renewable Signal

The unifying quantity across these literatures is fresh, learnable gradient signal per unit of human effort. Scraped text is a fixed stock: every token can be consumed once, repetition yields diminishing then negative returns [5][6], and the stock is forecast to be exhausted within the decade [3]. A verifiable environment inverts each of these properties. It is a generator, not a stock: a coding environment with a test suite, or a math environment with a checker, can score an unbounded number of fresh model attempts. The signal is self-renewing because it is produced by the model interacting with the checker, not extracted from a corpus. RLVE makes the consequence concrete — because each environment procedurally generates problems and adapts their difficulty to the policy, environments do not 'run out' the way a corpus does, which is why expanding their count keeps buying capability where added compute does not [14].

This reframes the role of human labor. In the RLHF paradigm, humans are in the inner loop, labeling preferences for every comparison — a per-token cost that does not amortize. In the RLVR paradigm, human labor is spent once to build the verifier (the test suite, the answer key, the adjudication rule), after which the environment grades arbitrarily many model rollouts for free. The marginal cost of an additional unit of training signal collapses from 'one human judgment' to 'one program execution.' This is the economic engine behind R1-Zero learning to reason from correctness alone [8] and behind RLVR generalizing across math and instruction-following in Tulu 3 [7]. Reported industry economics fit this shape: building an environment is a fixed, labor-intensive cost (per-task task-creation costs cited at $200-$2,000, with complex software-engineering tasks rarely up to ~$20K and full product replicas higher still), after which compute spent grading rollouts (~$2,400 per task in one report) is the variable line [16].

The corollary is a selection rule for which environments are worth building. Value accrues where (a) outcomes are objectively checkable and (b) the base model does not already latently hold the capability. Math and code score high on checkability but their substrate is largely public, so much of the RLVR gain there may be eliciting latent skill rather than teaching new skill — a reading reinforced by results showing RLVR can move some base models even under spurious or random rewards, model-dependently. The highest-value environments therefore sit in proprietary, under-documented verticals whose workflows nonetheless throw off objective pass/fail outcomes — clinical coding and claims adjudication, insurance, regulated-document review — where verifiability is present but the capability genuinely is not yet in the weights.

6. Implications for Data Strategy

The strategic consequence is a shift in what constitutes a defensible data asset. In the scraping era, the asset was a large, clean corpus; its value was eroded by deduplication, by the four-epoch repetition ceiling, and ultimately by exhaustion of the underlying stock [3][5]. In the verifiable-reward era, the asset is the environment — the paired (task distribution, verifier) that manufactures gradeable signal on demand. Unlike a corpus, an environment does not deplete with use and its signal does not decay with repetition, because each model attempt is novel, and because adaptive environments regenerate difficulty as the policy improves [14].

This favors a productized, recurring offering over a one-time data sale. A verifiable environment supports an effectively unbounded stream of training rollouts and a contamination-resistant evaluation stream from the same substrate (held-out tasks regenerated from the live workflow). The same asset thus serves both the training-data and evaluation-data markets — and the evaluation side carries a built-in renewal motive, since any static eval is contaminated once it leaks into a training set. The market is already structuring around this: third-party environment vendors report contract values in the 'six to seven figures per quarter,' and labs are building in-house while also buying — a make-and-buy pattern that signals an emerging supply chain rather than a one-off procurement [16].

It also sharpens vertical selection. The literature's biggest vertical wins share a template: proprietary or hard-to-assemble data crossed with an objective, checkable outcome — code with executable tests, math with answer keys [12][17]. Applying the verifiability lens, the most valuable environments to build are in verticals where (1) the underlying data is proprietary and absent from pretraining, and (2) workflow outcomes are deterministically gradeable. Clinical claims and procedure coding is the archetypal high-fit target: a private clinical record paired with a deterministic coding/claims pass-fail outcome (e.g. CDT/ICD/X12 adjudication). The pricing data underlines the moat: exclusive environment deals reportedly command a 4-5x premium over non-exclusive ones [15][16] — the market is already paying for exactly the proprietary exclusivity this thesis identifies as defensible.

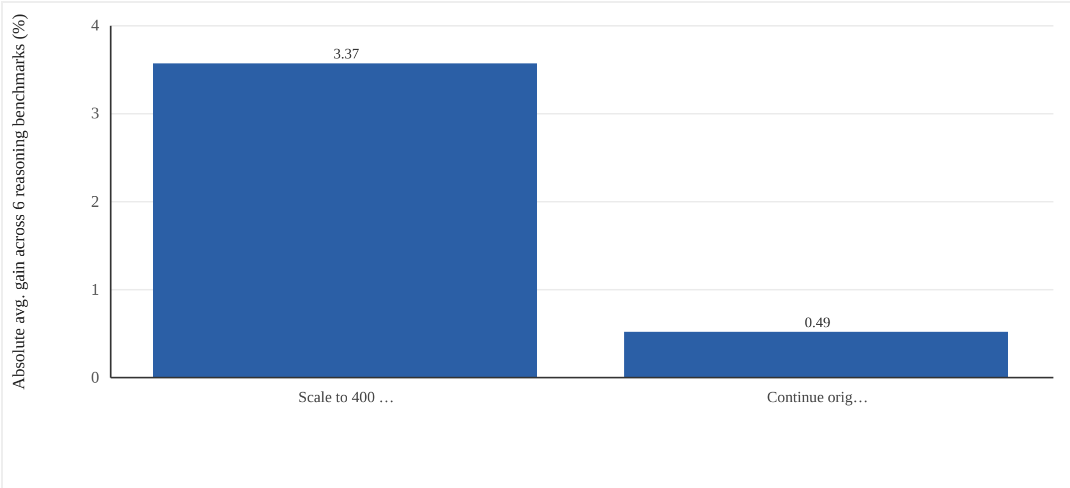

Figure 3. Environment scaling beats compute: RLVE controlled ablation. RLVE (Zeng et al., 2025) holds the base model fixed and varies the lever. Jointly training across 400 procedurally-generated verifiable environments (RLVE-Gym) yields a 3.37% absolute average improvement across six reasoning benchmarks; continuing the model's original RL training with over 3x more compute yields only 0.49%. This is the cleanest direct evidence that, post-data-wall, capability scales with the SUPPLY of verifiable environments rather than with compute on a fixed task distribution - the empirical core of the 'environments as a data asset' thesis. *Zeng et al., 'RLVE: Scaling Up RL for Language Models with Adaptive Verifiable Environments' (arXiv:2511.07317): 400 environments in RLVE-Gym, 1.5B base; 3.37% vs 0.49% (>3x compute) across six reasoning benchmarks. Verified against the paper's abstract. · Empirical data.*

7. Investment Thesis

The investable claim is timing plus structure, and it is no longer purely prospective — the market has begun to form around it. Timing: the data wall is a near-term, quantified forecast — exhaustion of public human text between 2026 and 2032, pulled toward 2027 under a 5x overtrain and as early as ~2025 under the most aggressive overtraining [3] [9]. The only demonstrated source of fresh signal past the wall is interaction with a verifiable environment, and that recipe (RLVR) became the core of frontier reasoning in the same window [7][8], with capability now scaling on environment supply rather than compute [14]. A buyer of training data in 2026 is structurally shifting budget from 'more text' (depleting, commoditizing) to 'more verifiable environments' (renewable, manufacturable).

Market formation, with numbers. As of late 2025, Anthropic was reported to be considering spending more than $1 billion on RL environments over the following year [15]. The adjacent data-labeling layer that environments are poised to succeed shows the scale of the prize: Surge AI reported $1.2B in revenue, Scale AI took a $14.3B investment from Meta (~49% stake, $29B valuation), and Mercor reached a $10B target valuation on a $450M run-rate [15]. Specialist environment startups (Mechanize, Prime Intellect, General Reasoning) are forming explicitly to be 'the Scale AI for environments,' with Mechanize reportedly offering $500K salaries to environment engineers [15]. The binding constraint that founders report is supply — the difficulty of scaling up task creation at the required quality level — i.e. demand exceeds the supply of high-quality verifiable environments [15]. That is the textbook setup for a data-asset business.

Structure: a verifiable-environment business has the economics investors prize. The marginal cost of an additional unit of training signal is a program execution, not a human label, so gross margins scale with usage rather than headcount. The asset compounds — a built verifier serves training, re-training at new compute budgets, and evaluation simultaneously, and the evaluation stream renews on a contamination clock. And the moat is the proprietary substrate: methods (GRPO, DAPO, PRMs, RLVR) are openly published and copied within months [13], but a proprietary vertical corpus paired with a domain-specific verifier is precisely the component competitors cannot assemble from the open web — which is why the market already prices exclusivity at a 4-5x premium [16].

The defensibility argument is the inverse of the data wall. The wall is bad for anyone whose asset is scraped text and good for anyone whose asset is a renewable, checkable environment in a vertical the open web underrepresents. The rare tails of the data manifold — specialized, regulated, proprietary workflows — are exactly where the marginal token is worth most and where redundancy-removal on the open web has nothing left to give. Concretely, the thesis underwrites a vertical-environment company: select verticals scoring high on both proprietary-data and verifiable-outcome axes; secure exclusive access to the real-data anchor; build the verifier once; and sell the resulting fresh training-and-eval signal as a recurring, compute-budget-indexed product. The dental and broader clinical claims/coding workflow — private clinical records adjudicated against deterministic code sets — is the worked high-fit example.

8. Limitations and Counterarguments

The exhaustion forecast is a projection, not a measurement. Epoch AI's own revision — from an earlier 'high-quality text exhausted by 2024' estimate to ~2028 — shows the window moves with methodology and with advances in data efficiency, multimodal data, and transfer from data-rich domains [3][9]. Synthetic data and improved data-efficiency could extend the runway — a 2025 study reaching the same loss asymptote with 5.17x less data under a properly regularized recipe is the most concrete standing counter-evidence on the efficiency front [22]; our thesis does not require the wall to arrive on a precise date, only that fresh human text is a depleting stock while verifiable environments are not.

RLVR's gains are not uniformly 'new capability.' Recent work shows RLVR can improve some base models even under spurious or random rewards, suggesting part of the effect is eliciting latent skill rather than teaching it — and that this is model-dependent. There is also a live dispute over whether RL expands the reasoning boundary or only sharpens what the base model already reaches (pass@k narrowing vs. expansion). This strengthens rather than weakens the vertical thesis — value concentrates where skill is genuinely absent — but it cautions against assuming every environment teaches; environments in domains saturated in pretraining may mostly surface existing ability.

The environment-scaling evidence is still thin and small-scale. RLVE's keystone ablation (3.37% vs 0.49%) is a single study at 1.5B parameters on procedurally-generated reasoning environments [14]; whether the 'environment count beats compute' result holds at frontier scale and in messy real-world verticals is an open extrapolation, not a settled law — though 2026 evidence is accumulating: Endless Terminals scales a fully autonomous pipeline to a large pool of synthetic terminal-agent environments with no human annotation and shows the resulting RL gains transfer to the human-curated Terminal-Bench 2.0 [20], extending the environment-supply lever from RLVE's 1.5B reasoning setting into frontier-style agentic tasks. The market figures in Section 7 are drawn from press reporting and vendor interviews, not audited financials, and should be read as directional indicators of a forming market rather than precise quantities.

Verifier design is itself a hard problem. Reward over-optimization (Goodharting) is well documented for learned rewards, and even programmatic verifiers can be gamed if the checkable proxy diverges from the true objective (e.g. tests that pass without a correct fix; SWE-bench audits have flagged underspecified tests). Building robust, non-hackable verifiers in messy real-world verticals — where 'correct' is adjudicated, not computed — is the central execution risk. Finally, proprietary vertical data carries access, licensing, and regulatory constraints (e.g. HIPAA, data-resale rights) that bound supply and must be cleared before an environment can be built or sold.

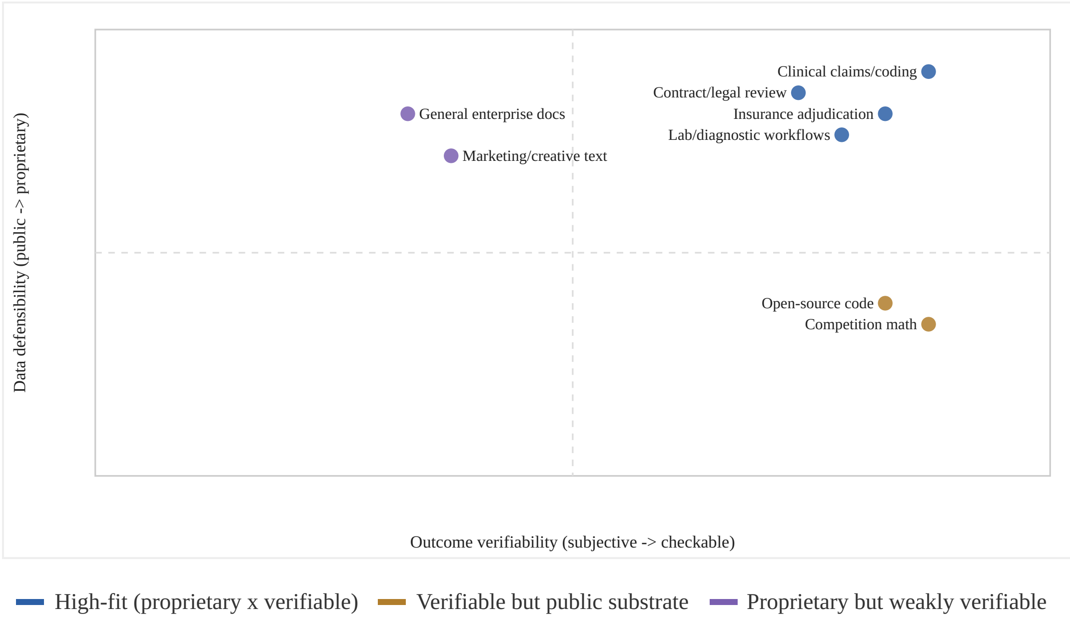

Figure 4. Strategic positioning: data substrate vs. outcome verifiability. A constructed strategic map of candidate verticals on two axes drawn from the synthesis: outcome verifiability (the renewable-signal axis) and data defensibility (the moat axis). The top-right quadrant - proprietary substrate crossed with checkable outcomes - is where verifiable-reward environments are both buildable and durably defensible, and is where the market's reported 4-5x exclusivity premium accrues. Math and code score high on verifiability but their public substrate thins the moat; many enterprise corpora are proprietary but lack a deterministic verifier. Axis positions are illustrative, not measured. *Constructed strategic analysis synthesizing the verifiability selection rule (Section 5) with vertical-win templates from arXiv:2310.06770 and arXiv:2411.04872, and the exclusivity-premium datum from TechCrunch (Sep 2025) and Epoch AI (2026) RL-environments reporting. · Constructed analysis.*

9. Conclusion

Three independent results — that public human text runs out in the late 2020s, that repeating it past a handful of epochs stops helping, and that repetition actively damages reasoning circuitry — converge on a conclusion none of them states alone: scaling on scraped text is ending as a source of fresh gradient signal. Over the same window, RLVR moved from niche to the core recipe behind frontier reasoning, the hardest open benchmarks are all verifiable, and a controlled ablation now shows capability scaling with the *supply of verifiable environments* rather than with compute. The synthesis is that the data wall is the economic reason verifiable-reward environments become a durable, manufacturable data asset class.

For a data-strategy and investment lens, the actionable claim is narrow and timely: the defensible asset of the next phase is not a corpus but an environment — a proprietary vertical substrate paired with an objective verifier that manufactures fresh training and evaluation signal on demand. The market has already begun to price this, with reported nine- and ten-figure commitments and an explicit 'Scale AI for environments' land-grab underway. Building such environments in real verticals with checkable outcomes is precisely the asset that sidesteps the data wall, and the literature — and now the market — say the timing is now.

INVESTMENT THESIS

The data wall is a near-term, quantified event: Epoch AI puts the effective stock of public human text at ~300T tokens (90% CI 100T-1000T) and projects full utilization between 2026 and 2032 - pulled to ~2027 under a 5x overtrain and as early as ~2025 under the most aggressive overtraining. Both obvious escapes are closed: repeating the corpus past ~4 epochs adds little and by ~16 essentially nothing [5], and repetition actively damages the reasoning circuitry [6]. The only demonstrated source of fresh gradient signal past the wall is interaction with a verifiable environment - and capability now scales on environment SUPPLY, not compute: RLVE's controlled ablation gets +3.37% from scaling to 400 environments versus +0.49% from >3x more compute [14]. The same recipe (RLVR) became the core of frontier reasoning in this window - DeepSeek-R1-Zero went 15.6%->71.0% on AIME from correctness alone, now Nature-peer-reviewed at a ~$294K RL training cost [8]. Crucially, the market is already pricing this: Anthropic reportedly weighed >$1B/yr on environments, environment engineers are bid to ~$500K, and exclusive deals command a 4-5x premium [15][16]. The investable consequence: the defensible asset is shifting from a corpus (depleting, commoditizing) to an environment (renewable, software-margin - marginal signal is a program execution, not a human label). The moat is the proprietary substrate, because methods are copied in months but a proprietary vertical corpus plus a domain verifier cannot be scraped. Own the real-data anchor in a vertical the open web underrepresents but whose outcomes are deterministically checkable; clinical claims/coding is the archetypal high-fit target. Timing is the edge - this market is forming exactly as the wall arrives.

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